UCC 103 PRINCIPLES OF COMPUTING
Chapter 9: Internet and E-mail

9.1. Internet

The Internet is a global system of interconnected computer networks that use the standard Internet protocol suite (often called TCP/IP, although not all applications use TCP) to serve billions of users worldwide.

It is a *network of networks* that consists of millions of private, public, academic, business, and government networks, of local to global scope, that are linked by a broad array of electronic, wireless and optical networking technologies. The Internet carries an extensive range of information resources and services, such as the inter-linked hypertext documents of the World Wide Web (WWW) and the infrastructure to support email.

9.2. Requirements for Connecting to the internet

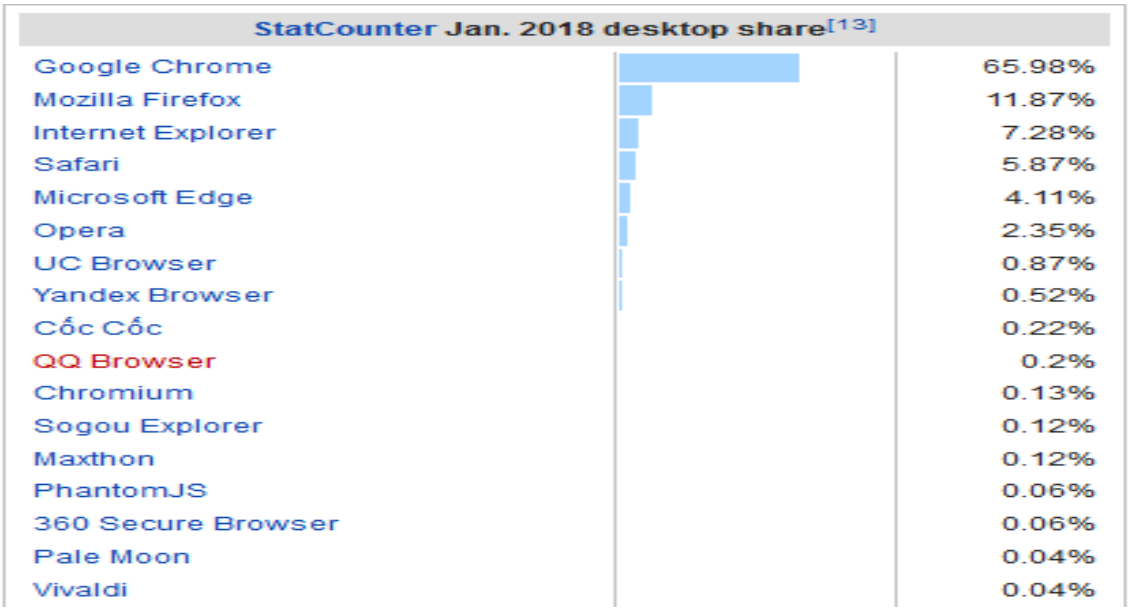
Requirements for connecting to the internet

1) **Computer or Electronic device**
2) **Internet service provider –** an internet service provider provides you with a connection to the internet and the software you will need to navigate.
3) **Telecommunication line** – a telephone line is required to connect you to the internet service provider.
4) **Modem** – a modem converts a digital signal received from a computer into an analogue signal that can be sent along ordinary telephone lines, and back to digital at the other end.
5) **Web browser –** a web browser is software used to view and download Web pages and various types of files such as text, graphics and video. Examples are Microsoft Internet Explorer or Mozilla Firefox, Google Chrome.

Basic internet browser features

1) Navigation buttons. Refresh button is used to go back and forward while **browsing**. ...
2) Refresh button. Refresh buttons is used to force web **browser** to reload webpage. ...
3) Stop button. ...
4) Home button. ...
5) Web **browser's** address bar. ...
6) Integrated search. ...
7) Tabbed **browsing**. ...
8) Bookmark buttons.
9) History

Browsers MarketShare



9.3. Ways of Connecting to the Internet

There are various type of connectivity to get hook on to Internet. They all can be broadly classified into following category. The choice of the connection methods depends on the Internet service providers chosen

1) Broad Band Connections
    a. Gate Way Access
    b. Leased Connections
    c. Digital Subscriber Line
    d. Cable Connections
2) Wireless connections
3) Dial-up Connections
4) Satellite
5) 3G and 4G Connections

9.3.1. Broad Band Internet Connection

Definition: The term broadband refers to any type of transmission technique that carries several data channels over a common wire.

a) **Gateway Access:** Gateway Access is also known as Level-One connection. It is the access to the Internet from a network, which is not on the Internet. The gateway allows the two different types of networks to "talk" to each other. But the users of the Gateway Internet have limited access to the Internet. They might not be able to use all the tools available on Internet. The local Internet Service Provider (ISP) normally defines this limitation.
b) **Leased Connection:** Leased connection is also known as direct Internet access or Level Three connection. It is the secure, dedicated and most expensive, level of Internet connection. With leased connection, your computer is dedicatedly and directly connected to the Internet using high-speed transmission lines. It is on-line twenty-four hours a day, seven days a week. Leased Internet connections are limited to large corporations and universities who could afford the cost.

c) **Digital Subscriber Line (DSL):** Digital Subscriber Line (DSL) is the newest technology being used for Internet access. DSL connects your home or office to the Internet through the same telephone wire that comes from the telephone pole on the street. DSL service, combines separate voice and data channels over a single telephone line. In DSL, voice fills the low end of the frequency spectrum and data fills the high end. Like ISDN, with DSL, user can make and receive telephone calls while connected simultaneously to the Internet.

> ***Integrated Services Digital Network (ISDN) is a set of communication standards for simultaneous digital transmission of voice, video, data, and other network services over the traditional circuits of the public switched telephone network.***

However, DSL service is limited in the distance that you can be from the provider's point of presence (POP). To use DSL you will need a DSL modem (also called a router), a Network Interface Card (NIC), and a telephone line. DSL is also relatively new technology that is just being introduced in many places.

d) **Cable Connection:** Cable service connects to the Internet **via cable TV**, although you do not necessarily need to have cable TV in order to get it. It uses a broadband connection and can be faster than both dial-up and DSL service; however, it is only available where cable TV is available.

9.3.2. Wireless Internet Connections

Wireless Internet, or wireless broadband is one of the newest Internet connection types. Instead of using telephone or cable networks for your Internet connection, you use radio frequency bands. Wireless Internet provides an always-on connection which can be accessed from anywhere as long as you geographically within a network coverage area. Wireless access is still considered to be relatively new, and it may be difficult to find a wireless service provider in some areas. It is typically more expensive and mainly available in metropolitan areas.

9.3.3. Dial-up Connection

Definition: Dial up networking technology provides PCs and other network devices access to a LAN or WAN via standard telephone lines.

'Dial-up' connection is also known as Level Two connection. This provides connection to Internet through a dial-up terminal connection. User's computer connect to the Internet by dialing a phone number (which is provided by your ISP) to connect to the ISP computer that provides the internet access, and connects to the network. The computer, which provides Internet access is known as 'Host' and the computer that receives the access, is 'Client' or 'Terminal'. The client computer uses modem to access a "host" and acts as if it is a terminal directly connected to that host. So this type of connection is also known as 'Remote Modem Access' connection. And the host to which the client gets connected is actually connected to the Internet by a full time connection (See Leased Connection).

In dial-up connection to Internet, Host carries all the command that are typed on a client machine and forward them to Internet. It also receives the data or information from the Internet on behalf of the 'Client' and passes it to them. The client computer acts as a 'dumb' terminal connected to remote host. Dial-up is an analog connection because data is sent over an analog, public telephone network. The modem converts received analog data to digital and vise

versa. Because dial-up access uses normal telephone lines the quality of the connection is not always good and data rates are limited.

9.3.4. Satellite Internet – Internet over Satellite (IoS)

Satellite Internet is a form of high-speed Internet service. Satellite Internet services utilize telecommunications satellites in Earth orbit to provide Internet access to consumers. Internet over Satellite (IoS) allows a user to access the Internet via a satellite that orbits the earth. A satellite is placed at a static point above the earth's surface, in a fixed position. Because of the enormous distances signals must travel from the earth up to the satellite and back again, IoS is slightly slower than high-speed terrestrial connections over copper or fiber optic cables.

Satellite Internet service covers areas where DSL and cable access is unavailable. Satellite offers less network bandwidth compared to DSL or cable. Typical Internet over Satellite connection speeds (standard IP services) average around 492 up to 512 Kbps.

9.3.5. 3G, 4G and 5G connections

These Generation connections are most commonly used with mobile phones, and it connects **wirelessly** through your ISP's network. However, these types of connections aren't always as fast as DSL or cable.

9.4. Internet / Network Cables

Network / Internet cables are used to connect and transfer information and data between storage area networks, switches, routers, and computers. The data flows through the network cables. Types of Cables commonly used includes;

1) Coaxial Cable. A plastic layer insulates the centre conductor and the braided metal. he cable can resist signal obstruction.
2) Shielded Twisted Pair (STP) Cable. It is mostly used in business installations. It is suitable for areas with risk to the electrical current and potential interference. Its external shield works as a ground. You can use the shielded twisted pair cable to expand the distance between cables.
3) Fibre Optic Cable. It consists of several layers of protective materials that surround a centre glass core. Fibre optic cable does not transmit electronic signals. Instead, it transmits light, removing the electrical obstruction problem. It is suitable for areas with electrical interference. You can use it to connect networks between buildings. ...
4) Unshielded Twisted Pair Cable. It is mostly used for computer networking and conventional telephone. It does not have internal shielding. It is suitable for workgroups because of its good performance and flexibility. It has an outer insulating jacket that protects the cable from damage or physical stress.

9.5. Security threats on the Internet

1) ***Malicious Software: Viruses, Worms, Trojan Horses, and Spyware –***
   Malicious Code; Malware, it includes a variety of threats such as viruses, worms, Trojan horses and bots.
   a) A virus is a computer program that has the ability to replicate or make copies of itself, and spread to other files. Worm is a malware that is designed to spread from computer to

computer. Viruses and worms spread across computers and networks by making copies of themselves, usually without the knowledge of the computer user.

- ✓ Computer virus fall in several major categories;
- ✓ Macro virus
- ✓ File-infecting virus
- ✓ Script viruses

b) A Trojan horse is a program that appears to be legitimate but actually contains another program or block of undesired malicious, destructive code, disguised and hidden in a block of desirable code. Trojans can be used to infect a computer with a virus.

c) A back-door Trojan is a program that allows a remote user or hacker to bypass the normal access controls of a computer and gain unauthorised control over it. Typically, a virus is used to place the back-door Trojan onto a computer, and once the computer is online, the person who sent the Trojan can run programs on the infected computer, access personal files, and modify and upload files.

d) A botnet (short for robots) is a group of infected, remotely-controlled computers. The hacker sends out a virus, Trojan or worm to ordinary computers. The virus, Trojan or worm gains access to the computer, usually through some malicious application that they are carrying. This in turn allows the hacker to gain full control of the now-infected computers. These computers can then be used to launch denial-of-service attacks, distribute spam emails and commit click fraud, identity theft and thefts of log-in details and credit card numbers.

2) ***Hackers and Cyber-vandalism***

A **hacker** is an individual who intends to gain unauthorized access to a computer system. The term **cracker** is typically used for hackers with criminal intent. Hackers spoof, or misrepresent themselves, by using fake e-mail addresses or masquerading as someone else. Hacker activities include:

- Theft of goods and services
- System damage
- Cyber-vandalism: The intentional disruption, defacement, or even destruction of a Web site or corporate information system.
- Spoofing: Hiding of the hackers true identities or email addresses, or redirecting a Web link to a different web site that benefits the hacker.
- Theft of proprietary information: A sniffer is an eavesdropping program that monitors network information and can enable hackers to steal proprietary information transmitting over the network.

3) **Spoofing**

To misrepresent oneself by using fake email addresses or masquerading as someone else. It is one of the methods hackers and criminals can use to gain improper or illegal access to computer systems. Spoofing is becoming a common way to steal financial information through fake Web sites. The spoofed site is almost a mirror image of the real site and unless the unsuspecting user examines the spoof closely, he/she may inadvertently give out important personal and financial information.

4) **Sniffing:**

A type of eavesdropping program that monitors information travelling over a network. It is a popular way to "grab" information as it passes over transmission lines whether they are hard-

wired or wireless. It is almost impossible to detect and encryption is about the only way to safeguard against it.

5) **Denial of Service (DOS) and Distributed Denial of Service (DDOS)*:***
   An attack on a Web site in which an attacker uses specialized software to send a flood of data packets to the targeted computer with the aim of overloading its resources. Due to the widespread availability of free intrusion tools and scripts and the overall interconnectivity on the internet, virtually anyone with minimal computer experience can mount a DOS attack.
   DDOS: involves using numerous computers to attack the targeted network from numerous launch points.
6) **Computer crime and cyber terrorism**
   Computer crime is any violations of criminal law that involve knowledge of computer technology for their penetration, investigation or prosecution. The computer can be an instrument of crime or a target of crime.
7) **Identity theft**
   With the growth of the internet and electronic commerce, identity theft has become a matter of concern. Identity theft is a crime in which an imposter obtains key pieces of personal information, such as debit or credit card details to impersonate someone else.

9.6. E-Mail (Electronic Mail)

E-mail or Electronic mail is a paperless method of sending messages, notes or letters from one person to another or even many people at the same time via the Internet. E-mail is very fast compared to the normal post. E-mail messages usually take only few seconds to arrive at their destination. One can send messages anytime of the day or night and it will get delivered immediately. You need not to wait for the post office to open and you don't have to get worried about holidays. It works 24 hours a day, seven days a week. What's more, the copy of the message you have sent will be available whenever you want to look at it - even in the middle of the night. You have the privilege of sending something extra even such as a file, graphics, images etc. along with your e-mail. The biggest advantage to using e-mail is that it is cheap, especially when sending messages to other states or countries and at the same time it can be delivered to a number of people around the world.

Although e-mail is faster and cheaper, it has many of the components of regular mail. It allows you to compose note, get the address of the recipient and send it. Once the mail is received and read, it can be forwarded, replied. One can even store it for later use, or delete. In e-mail even the sender can request for delivery receipt and read receipt from the recipient.

Features of E-mail:

1) One-to-one or one-to-many communications
2) Instant communications
3) Physical presence of recipient is not required
4) Most inexpensive mail service, 24-hours a day and seven days a week
5) Encourages informal communication

9.6.1. Components of an E-mail Address

As in the case of normal mail system, e-mail is also based upon the concept of a recipient address. The email address provides all of the information required to get a message to the recipient from anywhere in the world. The syntax for Emails is

mailbox_name@mailprovider

1) Mailbox is the recipient name of the mail on the destination computer, where finally the mail will be delivered.
2) @ ampersand – Internationally accepted symbol for all emails
3) Mail provider – the name of the Email service provider, ie the organization that is hosting the mail server

   Common providers include; yahoo.com, Hotmail.com, gmail.com etc

Example, john@hotmail.com
jane@gmail.com
otieno@hotmail.com

9.7. Connecting to E-Mail

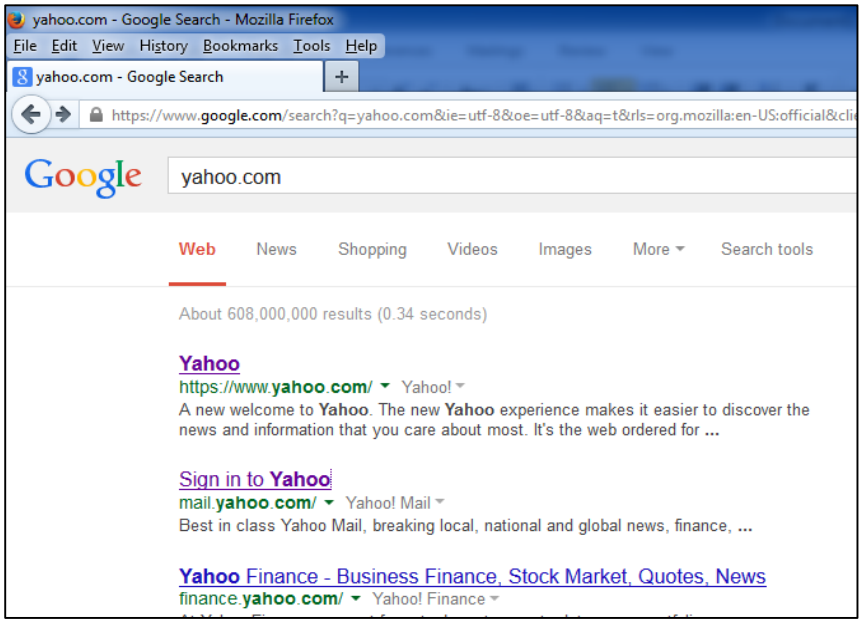
1. Double click the browser symbol to open it

Symbol	Browser Name
	Mozilla Fire-Fox
	Internet Explorer
	Google Chrome

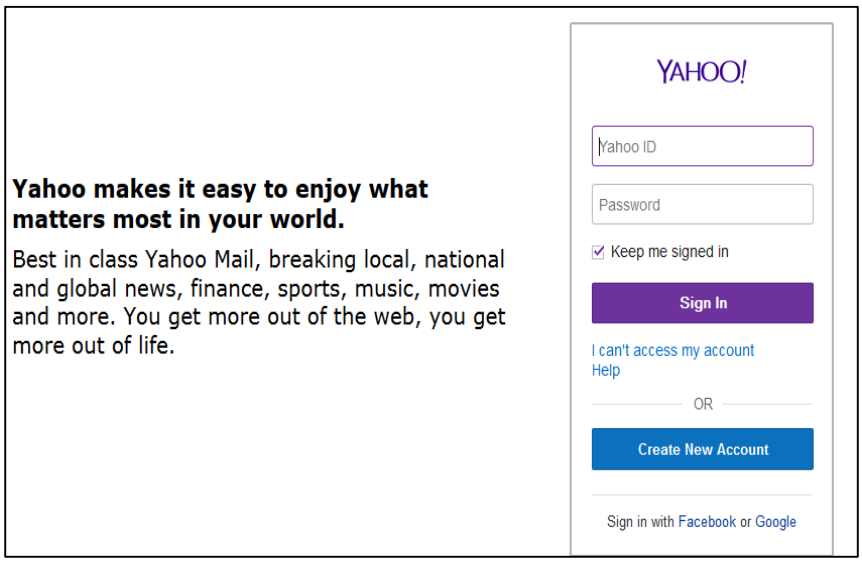
2. Type in the name if the email provider at the Search engine prompt and then click search. Example type in Yahoo.com to search for Yahoo mail provider.



3. Click Sign in to Yahoo options to open the Yahoo.com mail Sign in form.



4. At the Sign In form below



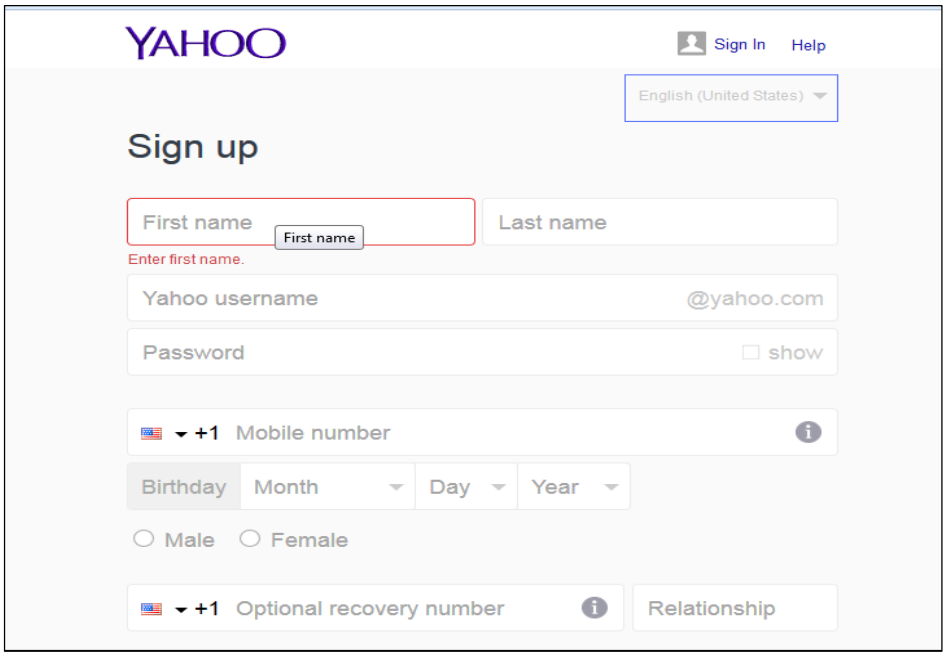
a). Have an Email Account

- Enter the email User Name
- Enter the Email Password
- Click Sign In Button

b). No Email Account

- Click the Create New Account Button

5. Enter the necessary information in the form provided use the Scroll bar to the right of the form to up and down the form.
   At the end of the form click the Create Account Button.



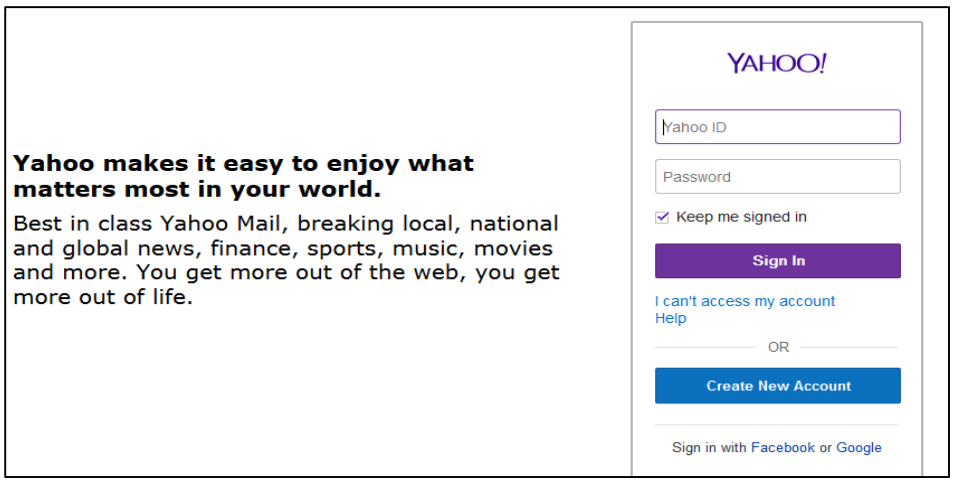
a) Enter your First name.
b) Enter your Surname
c) Enter the Preferred Username (Yahoo proposes Usernames choose any if appropriate)
d) Enter Password 8 – 32 Characters (Combine Characters, Numbers and Special Characters to make the password strong
e) Click the country field and select Kenya to insert the Kenya Flag Symbol and the **+254 code**



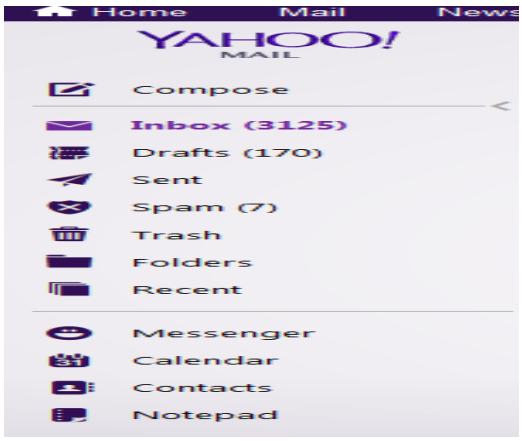
f) Enter your Mobile number (leave out the first Zero)
g) Enter the other required details appropriately.

9.8. Logging in to the created Email

Use the Username and password to log in into your Email Account/mailbox

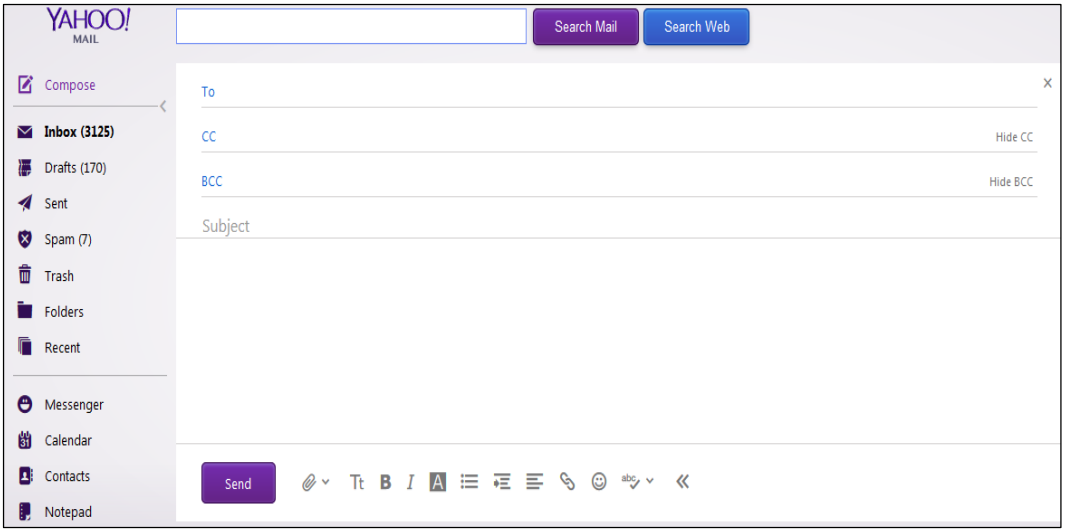


9.8.1. Mail Box Features



Feature	Roles / Description
Compose	Click to open a new template for a mail
Inbox	Click to open the inbox to view list of the mails stored there
Draft	Used to store draft emails created.
Sent	Stores a copy of ALL sent emails
Spam	Stores spam mails (mail flagged or tagged as dangerous mails – virus)
Trash	Store a copy of deleted mails
Folder	Creates a folder for storing mails
Recent	Shows recent mails

9.8.2. New Mail Template Features



To	Enter the Email(s) of the recipients. If many recipients, separate the emails by a comma(,). REQUIRED
CC	Enter the Email to receive a copy of the email. Copied emails are expected to take action on the received mail. OPTIONAL
BCC	Enter the email to receive a copy of the mail just to be informed, not action required. OPTIONAL

Subject	A word or statement to indicate the reason / title of the email. REQUIRED
	Click to send the email
	Click to attach a file to the email

9.9. Forwarding Mail, Replying Mail

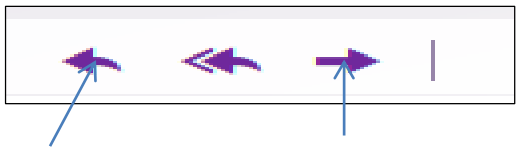
Emails can be forward to other recipients.
Select and click the relevant mail to open it

To forward:

- Click the forward button to open a new mail template with the current mail copied into it. Type in the recipient mail address to receive the mail
- Click send button

<u>To Reply</u>

- Click the reply to sender button
- Type in any comments as required
- Click Send button



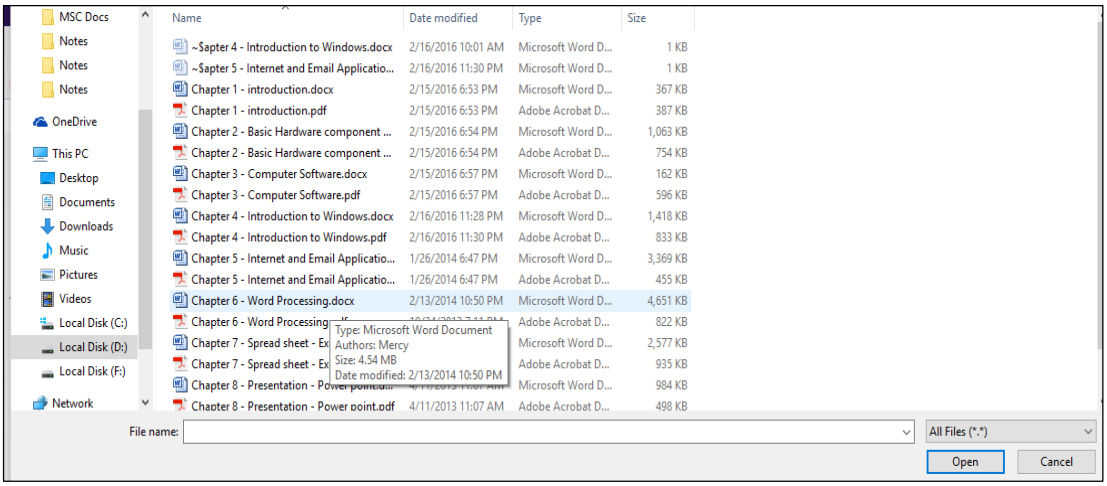
Reply to Sender Forward

9.10. Sending Mail with an attachment

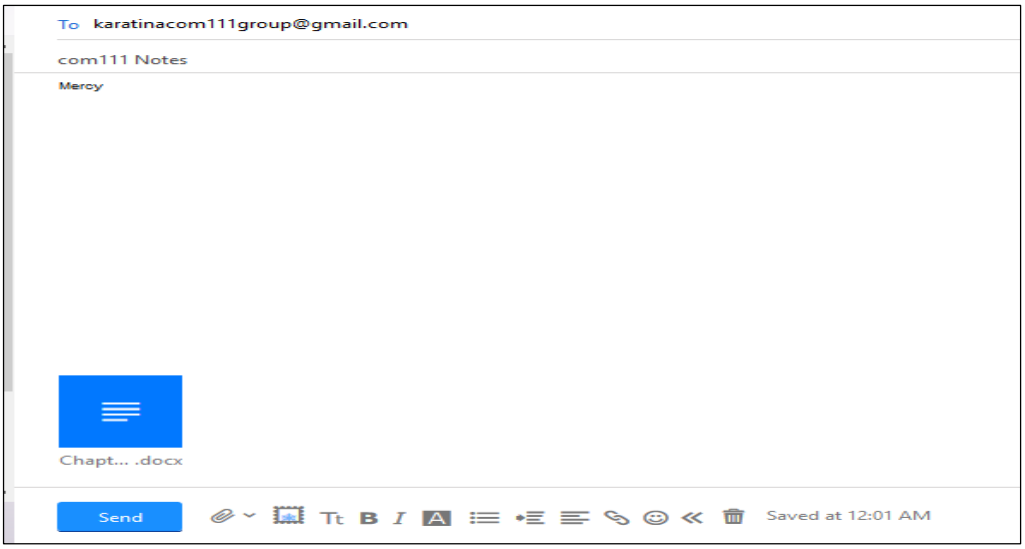
1) Create the mail
2) Select attachment icon next to SEND button



3) Select the file to attach from the respective folder and then Click Open button



4) Click Send Button once the attaching process for the attachment is complete.

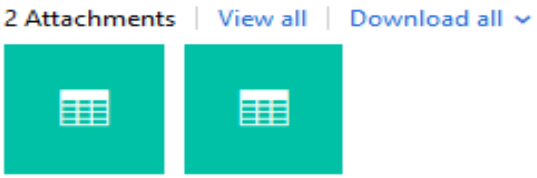


9.11. Opening an Attachment

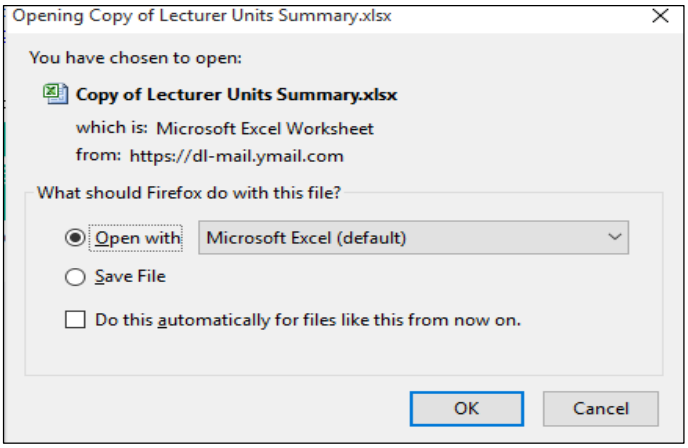
A mail with an attachment will have an Attachment icon (Staple pin) as shown below



1. Double-Click the email to open it. The attachment will be placed at the bottom of the letter



2. Double-click the attachment OR click Download all Link (to select all the attachments at the same time)



3. Select **Open with** radio button.
4. Select the respective application program to open the attachment with (click down arrow to open the application to open with selection box
   NB the symbol associated with the file will give you a clue on the application used to create it.
5. Click **OK button** to proceed to open the doc with the selected application ELSE Click **Cancel button** to exit from the window.